

Differential Equations in Geophysical Fluid Dynamics

XXV. Stratification and thermal wind

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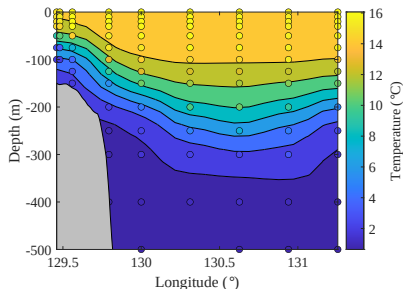
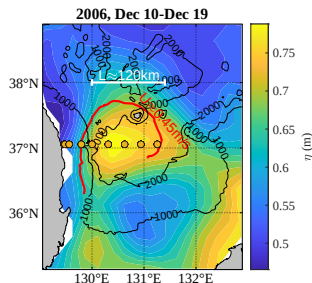
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Introduction

Cushman-Roisin and Beckers (2011) said:

“Two main ingredients distinguish the discipline from traditional fluid mechanics: the effects of rotation and those of stratification.”



Note that (typical) shallow water equations we used assume hydrostatic and constant density.

Introduction

Let us consider Boussinesq and hydrostatic ocean (considering density variation) with $Ro \ll 1$:

$$\begin{aligned} \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} - fu &= -\frac{1}{\rho_0} \frac{\partial P}{\partial x} \\ &+ \frac{\partial}{\partial x} \left(A_h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_h \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(A_z \frac{\partial u}{\partial z} \right) \end{aligned} \quad (1a)$$

$\vdots$

$$\begin{aligned} \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} &= -\frac{1}{\rho} \frac{\partial P}{\partial z} \\ &+ \frac{\partial}{\partial x} \left(A_h \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_h \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(A_z \frac{\partial u}{\partial z} \right) - g \end{aligned} \quad (1b)$$

Thermal wind equation

$$-fv = -\frac{1}{\rho_0} \frac{\partial P}{\partial x}, \quad fu = -\frac{1}{\rho_0} \frac{\partial P}{\partial y} \quad (2a)$$

$$0 = -\frac{1}{\rho} \frac{\partial P}{\partial z} - g \quad (2b)$$

Substituting (2b) into $\partial(2a)/\partial z$ yields

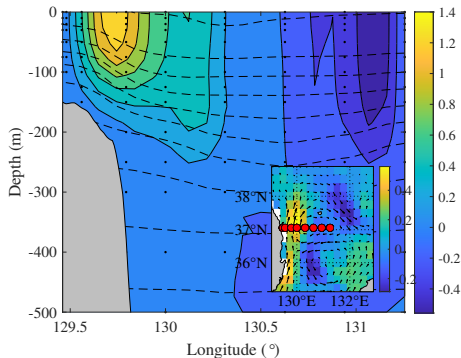
$$\frac{\partial v}{\partial z} = -\frac{g}{f\rho_0} \frac{\partial \rho}{\partial x}, \quad \frac{\partial u}{\partial z} = \frac{g}{f\rho_0} \frac{\partial \rho}{\partial y} \quad (3)$$

Geostrophic shear

that is referred to as **thermal wind equation**. No more Taylor-Proudman theorem! It has vertical gradient depending on lateral density gradient.

Thermal wind equation

By integrating (3) with a reference (e.g., level of no motion indicating depth where $u \approx 0$ and $v \approx 0$), geostrophic currents can be estimated from density fields.



Thermal wind component

$$-fv = -\frac{1}{\rho_0} \frac{\partial P}{\partial x} \quad (4)$$

$$0 = -\frac{1}{\rho} \frac{\partial P}{\partial z} - g \quad (5)$$

Let us decompose density ρ to constant component ρ_0 and variation component ρ' (so $\rho(z) = \rho_0 + \rho'(z)$). Approximated w -momentum equation (hydrostatic equation) can be simplified to

$$\begin{aligned} P(\eta) - P(z) &= -g \int_z^\eta \rho(z) dz \\ &\approx -g \left(\rho_0(\eta - z) + \int_z^0 \rho'(z) dz \right) \end{aligned} \quad (6)$$

$$\therefore P \approx g \left(\rho_0(\eta - z) + \int_z^0 \rho'(z) dz \right)$$

Thermal wind component

Substituting (6) into (4) yields geostrophic current considering stratification (vertical density variation):

$$v(z) = \underbrace{\frac{g}{f} \frac{\partial \eta}{\partial x}}_{\text{Barotropic component}} + \underbrace{\frac{g}{f \rho_0} \left(\int_z^0 \frac{\partial \rho'}{\partial x} dz \right)}_{\text{Baroclinic (thermal wind) component}} \quad (7)$$

Buoyancy shutdown

Let us consider generalized Ekman problem:

$$-fv = -\frac{1}{\rho_0} \frac{\partial P}{\partial x} + A_z \frac{\partial^2 u}{\partial z^2} \quad (8a)$$

$$fu = -\frac{1}{\rho_0} \frac{\partial P}{\partial y} + A_z \frac{\partial^2 v}{\partial z^2} \quad (8b)$$

$$0 = -\frac{1}{\rho} \frac{\partial P}{\partial z} - g. \quad (8c)$$

Using complex coordinate, momentum equations can be rewritten as

$$fi\vec{u} = -\frac{1}{\rho_0} \frac{\partial P}{\partial \vec{n}} + A_z \frac{\partial^2 \vec{u}}{\partial z^2} \quad (9)$$

For simplicity, consider constant density gradient:

$$fi\vec{u} = -\left(g \frac{\partial \eta}{\partial x} - \frac{g}{\rho_0} \left(\frac{\partial \rho'}{\partial x}\right) z\right) + A_z \frac{\partial^2 \vec{u}}{\partial z^2} \quad (10)$$

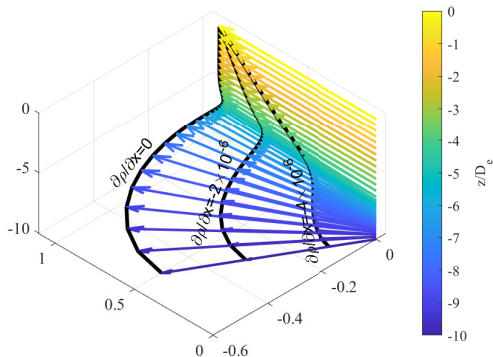
Buoyancy shutdown

Approximated solution using $1 \ll h/D_e$ is given by

$$\vec{u} \approx - \left(\frac{g}{f} \frac{\partial \eta}{\partial x} + \frac{gh}{f \rho_0} \frac{\partial \rho'}{\partial x} \right) \frac{1+i}{f D_e / \gamma} e^{-j(z+h)} + \vec{u}_p \quad (11)$$

where

$$\vec{u}_p = i \left(\frac{g}{f} \frac{\partial \eta}{\partial x} - \frac{g}{f \rho_0} \frac{\partial \rho'}{\partial x} z \right) \quad (12)$$



References I



Cushman-Roisin, Benoit and Jean-Marie Beckers (2011).
*Introduction to geophysical fluid dynamics: physical and
numerical aspects*. Vol. 101. Academic press.